Phase plane portraits

Draw solution curves for system $\underline{X}' = \underline{A}\underline{X}$ $(A_{2\times 2})$

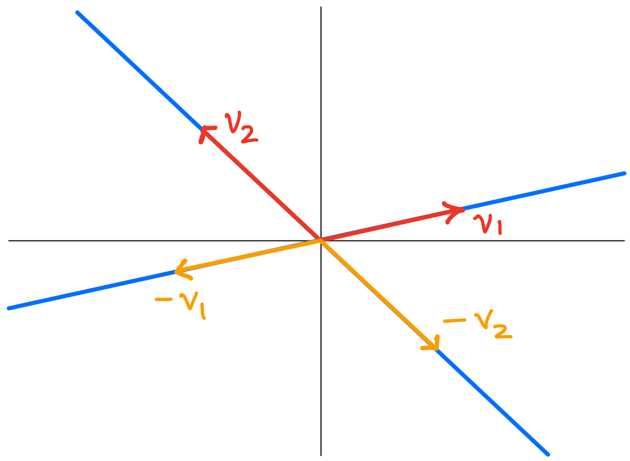
Cases where gen solution has form

$$\underline{X}(t) = c_1 e^{\lambda_1 t}\underline{v_1} + c_2 e^{\lambda_2 t}\underline{v_2}. \quad (c_1\underline{X}_1 + c_2\underline{X}_2)$$

note $\underline{X}(0) = c_1\underline{v_1} + c_2\underline{v_2}$ (initial position)

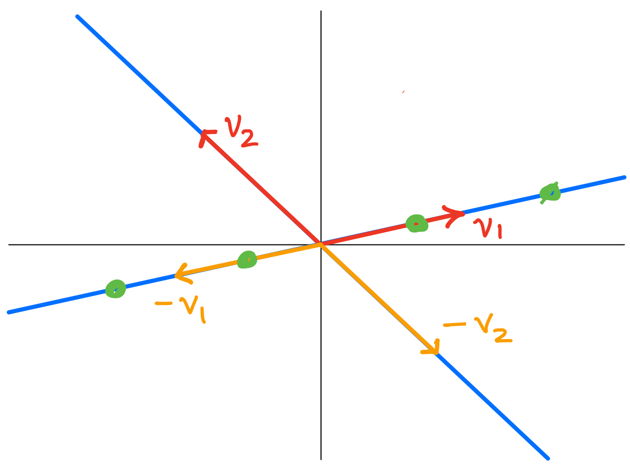
we assume t can range over all of $(-\infty, \infty)$

Case: repeated $\lambda \in \mathbb{R}$; no defect.

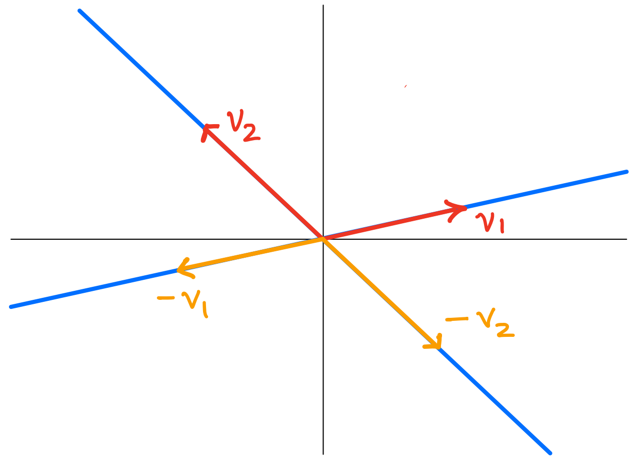


- $\lambda < 0 \longrightarrow$
- $\lambda > 0 \longrightarrow$
- $\lambda = 0 \longrightarrow$

Case: $\lambda_1 = 0$, $\lambda_2 \neq 0$, $\lambda_2 \in \mathbb{R}$.



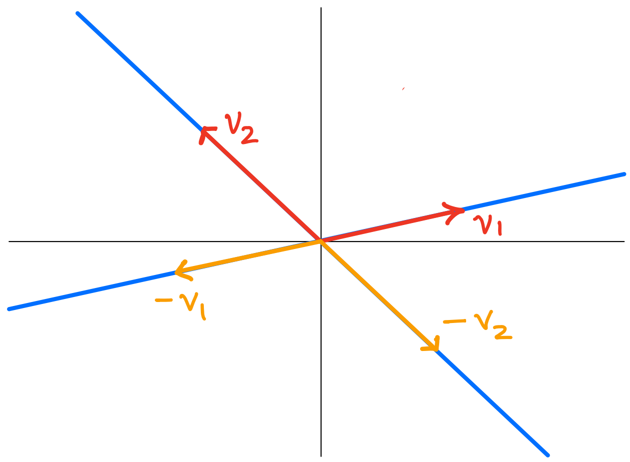
Case : $\lambda_1, \lambda_2 \in \mathbb{R}$, $\lambda_1 \neq \lambda_2$, same sign



"improper nodal source/sink"

Ex : $0 < \lambda_1 < \lambda_2$

Case $\lambda_1 \neq \lambda_2$, $\lambda_1 < 0 < \lambda_2$



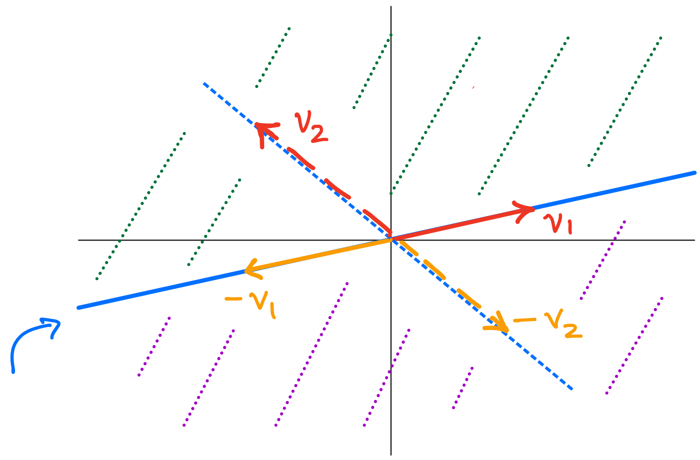
"saddle point"

Ex : $\lambda_1 < 0 < \lambda_2$

Cases of repeated λ with defect.

v_1 is genuine eigenvector

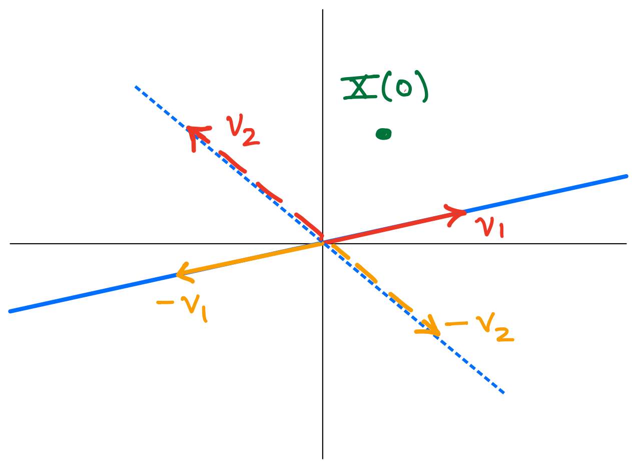
v_2 is generalized " " (2-step),



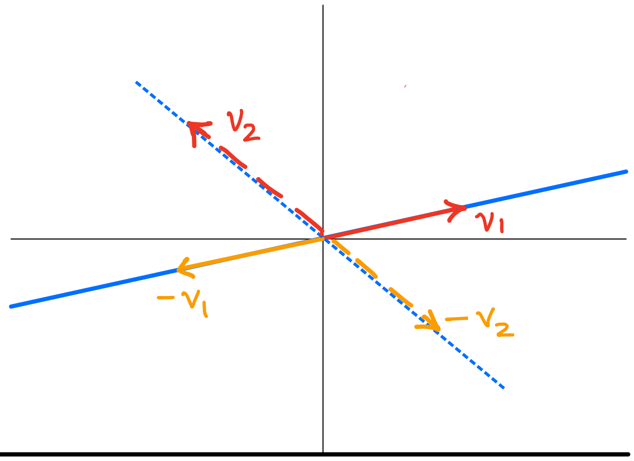
trajectory over time

ex: $\lambda > 0$, $c_1 = 1$, $c_2 = 1/2$

$$\underline{X}(t) = \left(1 + \frac{t}{2}\right) e^{\lambda t} \underline{v_1} + \frac{1}{2} e^{\lambda t} \underline{v_2}$$



If $\lambda = 0$ is repeated, defect > 0,



Case of complex conjugate pair $\lambda = p \pm qi$

Say $\underline{v}$ is eigenvector for $\lambda = p + qi$, and

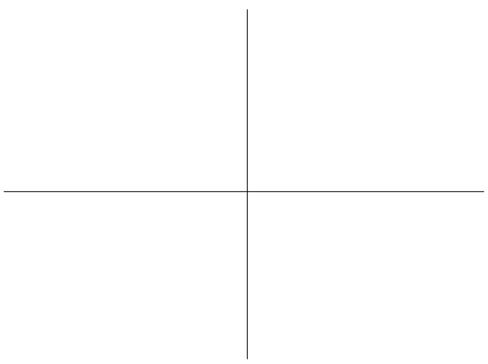
$\underline{v} = \underline{a} + \underline{b}\, i$ (real & imaginary parts)

Then basis solutions are

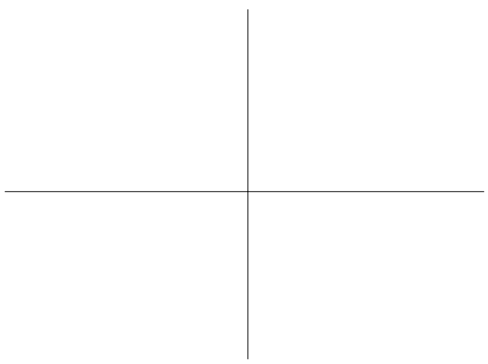
$$\underline{X}_1 = \mathrm{Re}\left(e^{\lambda t}\underline{v}\right) = \cdots =$$

$$\underline{X}_2 = \mathrm{Im}\left(e^{\lambda t}\underline{v}\right) = \cdots =$$

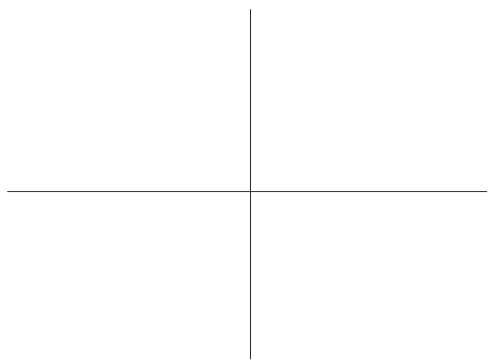
gen solution $\underline{X} = c_1 \underline{X}_1 + c_2 \underline{X}_2$



$p < 0$



$p = 0$



$p > 0$